Magnetic anomaly near the center of the Vredefort structure: Implications for impact-related magnetic signatures

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ABSTRACT

A strong magnetic anomaly near the center of the ancient and deeply eroded Vredefort structure is attributed to remanent magnetization caused by a large meteorite impact at ~2.0 Ga. The rocks underlying the anomaly are Archean gneisses thought to represent mid-crustal depths that were uplifted to the surface during the postulated impact event. Measurements of the remanent magnetization of the basement rocks yielded consistent vectors of declination = 25°, inclination = 56°, k = 18, α = 16 that correspond to the paleomagnetic pole position at time of impact. Petrologic studies indicate that during impact, large volumes of rock were heated enough to cause thermal remagnetization in the ambient field. Thermal effects of all large impacts on cratons may induce a remanent magnetization of sufficient intensity to cause anomalies in the geomagnetic field that are detectable even by satellites.

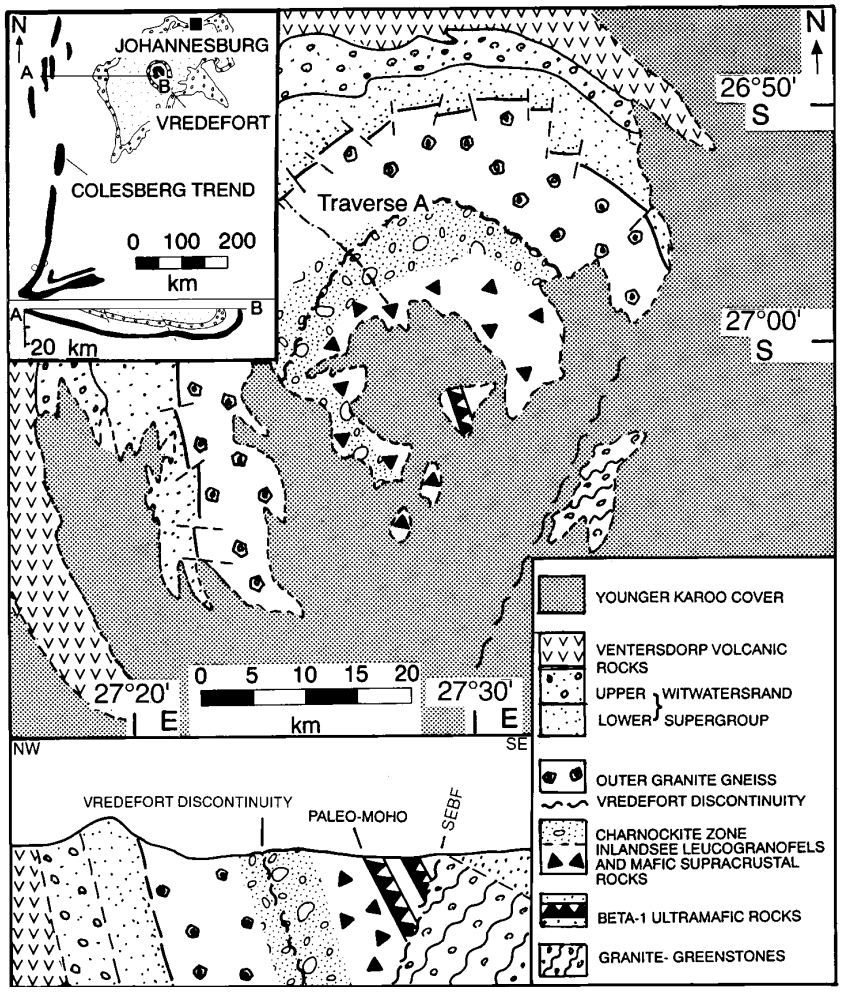


Figure 1. Geologic map of Vredefort structure and northwest-southeast section across it (after Hart et al., 1990). SEBF = southeast boundary fault. Horizontal scale of cross section same as map; vertical scale exaggerated by 30%. Position of traverse A is indicated by dot-dash line. Upper left inset shows Colesberg trend in relation to Vredefort structure.

INTRODUCTION

The origin of the Vredefort structure in South Africa has been debated for several decades. Although it is widely regarded as having been caused by impact (e.g., Grieve et al., 1990), there are many aspects of the structure that do not conform to those of known impact craters (Reimold, 1990). The morphology of this structure (Fig. 1) resembles that expected of the deep central zone of a large complex crater: current estimates suggest an original diameter of ~300 km (Therriault et al., 1993). The margin of the remnant structure consists of a semicircular series of ridges of Precambrian strata, surrounding a central flat area consisting mainly of Archean granitic basement. Hart et al. (1990) demonstrated that, in the exposed northwestern sector of the crater, the rim strata and the basement rocks all dip steeply inward toward the center and that the metamorphic grade increases toward the core of the structure. They postulated that almost the entire cross section (~36 km) of crust is exposed along a radial traverse from the outer margin to the core of the structure (Fig. 1).

Aeromagnetic images over the structure (Corner et al., 1990) show strong, well-defined concentric patterns (Fig. 2). In the rim, the patterns reflect the different sedimentary strata. Although the concentric patterns continue into the basement, the causes of variation of the magnetic field in the granitic rocks are less obvious. About halfway into the basement there is a prominent negative magnetic anomaly that extends in a broad semicircular belt ~2 to 4 km wide around most of the basement core. This anomaly is approximately centered on the Vredefort discontinuity (Hart et al., 1990), a mid-crustal tectonic boundary that separates upper crust in amphibolite facies from lower crustal granulites (see Figs. 1 and 2). The rocks that underlie the anomaly are Ar-

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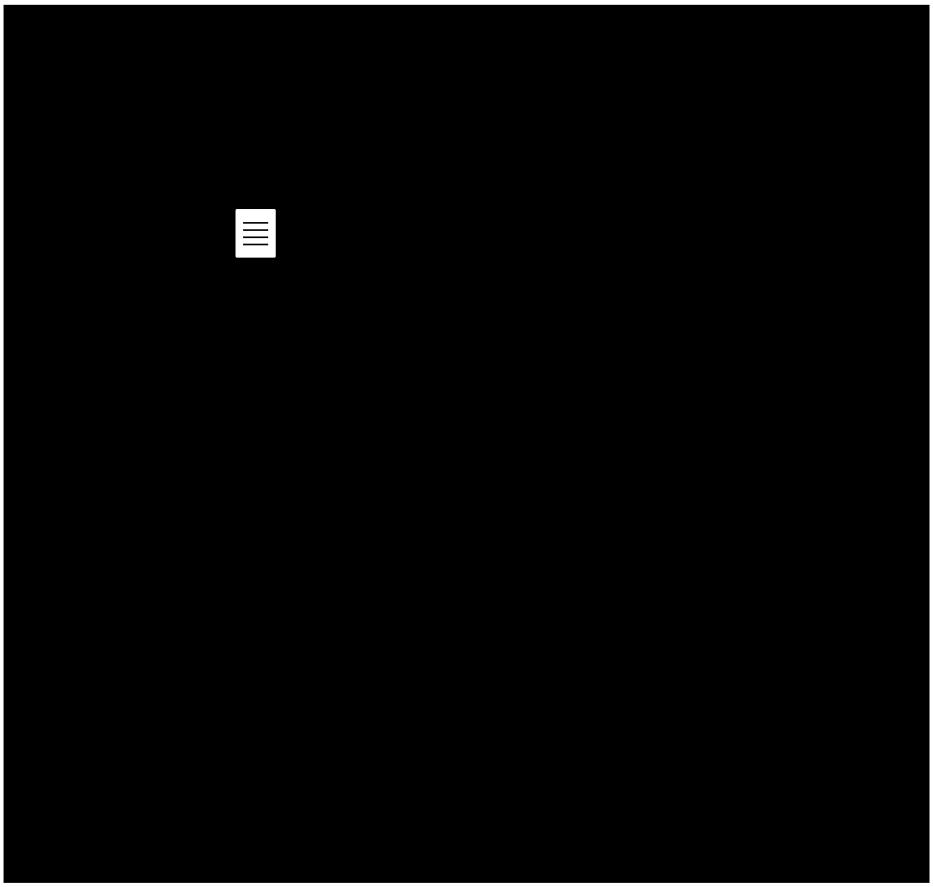


Figure 2. Map showing relation between lithologies and total aeromagnetic field across Vredefort basement. Circles show localities of paleomagnetic survey sites (after Jackson, 1982; Table 1).

chean gneisses and do not have unusually high magnetite content; further, there is no gravity signature indicative of dense Fe-rich rocks at depth.

Corner et al. (1990) speculated that the magnetic anomaly observed at Vredefort correlates with a prominent magnetic anomaly, termed the Colesberg trend, that generally follows the southern and western margins of the Kaapvaal craton (see Fig. 1). These authors interpreted the Colesburg trend as a regional mid-crustal layer of magnetite-rich granite that extends beneath the Witwatersrand basin and was uplifted at Vredefort during the 2.0 Ga event. However, Drennan et al. (1990) have investigated the rocks from the Colesberg trend in relation to the basement rocks at Vredefort and have provided petrographic, geochemical, and isotopic age data indicating that the two are significantly different. In this study we provide further petrological and petrophysical evidence indicating that the anomaly near the center of Vredefort is not related to a preexisting deep-seated magnetite-rich layer in the crust, but rather is caused by remanent magnetization acquired during or after uplift following a thermal event associated with the 2.0 Ga impact.

PALEOMAGNETISM

Table 1 summarizes the results of a ground magnetic and paleomagnetic survey from 16 sites (Fig. 2) throughout the granite basement (Jackson, 1982). Two sites gave inconsistent results and four yielded results similar to the present Earth's field; the cause of this component is not known. The directions from six sites (1, 8, 9, 10, 13, and 14) are in a group with a northern declination (Dec) and positive inclination (Inc) (designated group A), which averages Dec = 25°, Inc = 56°, $k = 18$, $\alpha = 16$. Of the remaining four sites, 5 is intermediate between this direction and the present Earth's field, whereas sites 3, 7, and 19 are reversed. Four of the six group A sites (and site 5) are located approximately in the region that corresponds to the annular aeromagnetic anomaly.

The direction obtained for the group A samples is very similar to the average direction obtained from bronzite granophyre dikes that intrude into the northern parts of the basement (Hargraves, 1970). The granophyre dikes, which are interpreted by some (Dence, 1971; French and Nielsen, 1990) to be impact melts, are dated at 2002 ± 52 Ma (Walraven et al., 1990). Thus, it would appear that large parts of the basement at Vredefort were remagnetized at ~2.0 Ga.

The calculated pole position (21.7°N, 49.0°E) of the group A samples is also close to that obtained for gabbros from the ~2.0 Ga Bushveld complex (Hattingh, 1991), which is situated ~250 km to the north of Vredefort. The similarity of the magnetic direction of the Vredefort basement to other ~2.0 Ga rocks in the region suggests that the basement rocks at Vredefort must have been remagnetized during or after the central core was uplifted and overturned.

MAGNETIC PROFILES ACROSS THE BASEMENT

To expand on the work of Jackson (1982), we carried out a study of the magnetic properties of 46 samples taken along a traverse across the magnetic anomaly in the northwest section of the structure (see Fig. 1). Two to four cores were cut from each sample for bulk susceptibility and natural remanent magnetization (NRM) measurements. The susceptibility was measured by using a Digico susceptibility bridge and the NRM on a Digico spinner magnetometer. The magnetite content in these rocks was calculated from the major element chemistry given by Hart et al. (1990) by using CIPW norm calculations. These data are listed in Table 2 and plotted in Figure 3.

The aeromagnetic profile across the basement along traverse A (see Fig. 3A) shows a pronounced, nearly symmetrical negative magnetic anomaly of ~1000 nT, approximately centered on the Vredefort discontinuity. The susceptibility along traverse A ranges from ~0.01 to 2.00×10^{-3} emu/($cm^3 \cdot Oe$) (cgs units), consistent with the modest magnetite contents (<2%, see Fig. 3B). The intensity of the NRM (Jn, Fig. 3D) across the basement is variable but exceeds 20×10^{-3} emu/cm^3 (cgs units) in the vicinity of the Vredefort discontinuity. The values are lower toward the center of the structure as well as toward the outer parts of the basement.

The average Q ratios (Q = intensity of NRM/susceptibility × local Earth's field) across the basement (Fig. 3E) are extremely high (~52), indicating that remanence rather than induced magnetization dominates the total magnetic field. Jackson (1982) has demonstrated that with a remanence orientation of Dec = 25°, Inc = 56°, the magnetic anomaly can be accurately modeled by using magnetic properties comparable to those measured along traverse A.

THERMAL METAMORPHISM IN THE VREDEFORT BASEMENT

Thermal metamorphic features, e.g., mineralogical and textural changes and the local

TABLE 1. PALEOMAGNETIC DATA

Site no.	n	ODF (mT)	Dec (o)	Inc (o)	k	α	J n ($\times 10^{-3}$cgs)	χ ($\times 10^{-3}$cgs)	Q*
1	8	60	21	+68	34	9.6	0.18	0.39	1.6
2		results scattered					10.52	0.57	66.9
3	8	85	183	+11	20	12.5	5.07	0.89	18.6
4		results scattered					24.36	1.90	46.7
5	9	100	30	-10	16	13.4	6.28	0.52	37.2
6	7	40	00	-49	66	7.4	14.75	0.47	104.2
7	6	80	233	-69	9	23.5	10.40	0.40	84.0
8	8	90	43	+45	17	14.0	18.92	3.13	20.9
9	7	40	358	+45	30	11.1	25.78	2.00	45.1
10	8	40	10	+78	24	11.5	5.30	0.60	30.7
11	6	40	19	-80	70	8.0	5.66	0.04	500.0
12	8	50	357	-67	35	9.5	0.32	0.02	54.6
13	8	40	54	+44	14	15.2	1.39	0.67	7.0
14	8	60	13	+45	405	2.7	131.79	1.69	27.4
15	7	30	290	-50	13	17.5	1.72	0.41	14.3
19	8	40	195	-25	9	19.9	-37.69	1.77	71.4

Note: Data from Jackson, 1982; About 68 cores were drilled in the field and oriented by sun and magnetic compass. After measurement of natural remanent magnetism (NRM) and susceptibility (two specimens per core), stepwise alternating field (AF) demagnetization was performed on one specimen from each site, and an optimum AF to isolate a stable end point was selected. All other specimens from each site were then treated in this field. n = number of specimens; ODF = optimum demagnetizing field; Dec = declination; Inc = inclination; k = fisher's k; α = 95% confidence cone; Jn = intenity of NRM x 10^{-3} cgs; χ = susceptibility x 10^{-3} cgs; Q = Koenigsberger ratio; *local field measured at each site varies from 23 370 to 30 800 nT.

TABLE 2. MAGNETIC PROPERTIES OF SAMPLES ALONG TRAVERSE A

Sample no.	Distance from collar (km)	magnetite (%)	Jn ($\times 10^{-3}$ cgs)	χ ($\times 10^{-3}$ cgs)	Q*
A1	0.80	0.59	0.09	0.22	1.46
A2	1.40	0.54	3.25	0.10	96.91
A3	2.00	0.46	5.83	0.14	152.97
A4	2.30	0.46	1.94	0.21	30.47
A5	2.45	0.55	1.14	0.17	21.00
A6	3.10	0.53	1.43	0.20	23.69
A7	3.50	0.49	10.01	0.34	99.35
A8	3.80	0.33	5.72	0.15	127.15
A9	4.10	1.34	1 3.66	0.71	64.15
A10	4.50	1.21	21.61	0.70	103.42
A11	5.90	0.38	18.96	0.55	115.31
A12	5.15	0.50	5.15	0.31	61.98
A13	6.00	1.08	8.01	0.88	30.68
A14	6.50	1.58	22.23	0.43	186.40
A15	6.60	1.93	36.81	0.75	170.44
A16	6.62	0.52	0.62	0.26	7.98
A17	6.65	1.63	18.85	0.86	73.17
A18	6.75	1.70	3.10	0.89	10.86
A19	6.80	0.98	0.38	0.42	2.93
A20	6.90	1.91	6.59	1.13	19.37
----------Vredefort discontinuity----------					
A21	7.00	1.90	3.27	0.88	12.45
A22	7.05	1.42	1.59	0.26	21.12
A23	7.10	1.21	4.56	0.53	28.66
A24	7.20	0.51	2.04	0.20	34.91
A25	7.30	3.17	3.75	0.49	26.74
A26	7.40	1.06	0.95	0.21	13.96
A27	7.45	1.04	0.13	0.42	1.04
A28	7.50	1.22	1.59	0.82	22.83
A29	7.52	0.76	2.49	0.18	63.89
A30	7.80	0.97	2.94	0.28	35.24
A31	8.00	0.85	1.36	0.80	5.89
A32	8.10	0.46	12.82	0.40	32.04
A33	8.25	2.53	24.28	0.87	93.34
A34	8.50	0.67	12.84	0.28	154.14
A35	8.65	0.70	1.65	0.28	23.41
A36	8.70	2.03	2.84	2.00	43.14
A37	9.00	1.11	0.62	0.33	6.11
A38	9.10	0.92	5.45	0.32	53.53
A39	9.20	1.14	1.27	0.51	13.47
A40	9.25	0.20	0.25	0.01	128.08
A41	9.30	1.80	12.25	0.74	54.93
A42	9.50	1.65	1.40	0.06	75.31
A43	9.80	1.65	9.11	0.31	100.22
A44	10.40	1.06	0.76	0.31	8.06
A45	11.70	0.88	0.35	0.15	7.77
A46	11.90	0.48	0.10	0.08	4.31
A47	16.00	1.82	0.22	0.55	1.33

* Ambient field assumed to be 30 000 nT.

formation of anatectic melt rocks (Schreyer, 1983; Hart et al., 1991), have been cited as evidence for an ~2.0 Ga thermal event in the basement rocks. Schreyer (1983) estimated conditions ranging from ~500 °C at the edge to ~900 °C (at 2 to 4 kbar) in the center and further suggested that there is evidence that the thermal metamorphism in the core of the structure started before but outlasted the shock metamorphism characteristic of these rocks (Grieve et al., 1990). These findings imply that the shock metamorphism was superimposed on an ongoing thermal metamorphic event. The coincidence in time and space of the localized magmatism and the shock metamorphism was interpreted by Schreyer (1983) as indicative of an endogenous origin for the Vredefort structure. In contrast, Hart et al. (1991) argued that the thermal metamorphism in the core region of the structure clearly postdates the shock metamorphism and could be related to a single event such as an impact. Whatever the duration of the thermal event, there can be no doubt that the temperatures achieved in the Vredefort basement at ~2.0 Ga would have been sufficient to completely remagnetize all the basement rocks in the core.

CAUSE OF THE MAGNETIC ANOMALY

There are several aspects of the magnetic anomaly that militate against the premise that a preexisting magnetite-rich granitic layer in the crust was turned up during the 2.0 Ga event. Not the least of these is that the rocks that correspond to the anomaly do not contain anomalous amounts of magnetite. The magnetic anomaly is negative; therefore, remanence, rather than high susceptibility, should be the dominant factor, a conclusion confirmed by the high Q ratios in the basement. Further, the paleomagnetic data indicate that the basement rocks were remagnetized after they were uplifted to the surface.

It is possible that the anomaly is related to a regional thermal event such as the intrusion of the Bushveld complex. However, the anomaly and the thermal metamorphism are centered on the Vredefort structure, and there are several lines of evidence that relate the thermal metamorphism in the Vredefort basement to a single heat pulse, interpreted by Hart et al. (1991) to be the result of a meteorite impact. It is our premise that the same heat pulse remagnetized the basement rocks in the core of the structure at ~2.0 Ga.

CONCLUSIONS

Magnetic signatures related to impact events are rare (for examples, see Halls, 1979), and large aeromagnetic anomalies associated with meteorite craters are thought to be unrelated to the impact event (Pilkington and Grieve, 1992). However, most other craters studied are small and comparatively young; Vredefort is one of the largest and oldest (subsequent erosion estimated at 8–10 km) impact structures on Earth. We contend that the data presented for the magnetic anomaly in the Vredefort basement are consistent with impact-related thermal remanent magnetization. Although the very strong remanence is confined to rocks close to the Vredefort discontinuity, the evidence suggests that the entire basement may have been remagnetized in the Earth's ambient field at the time of impact

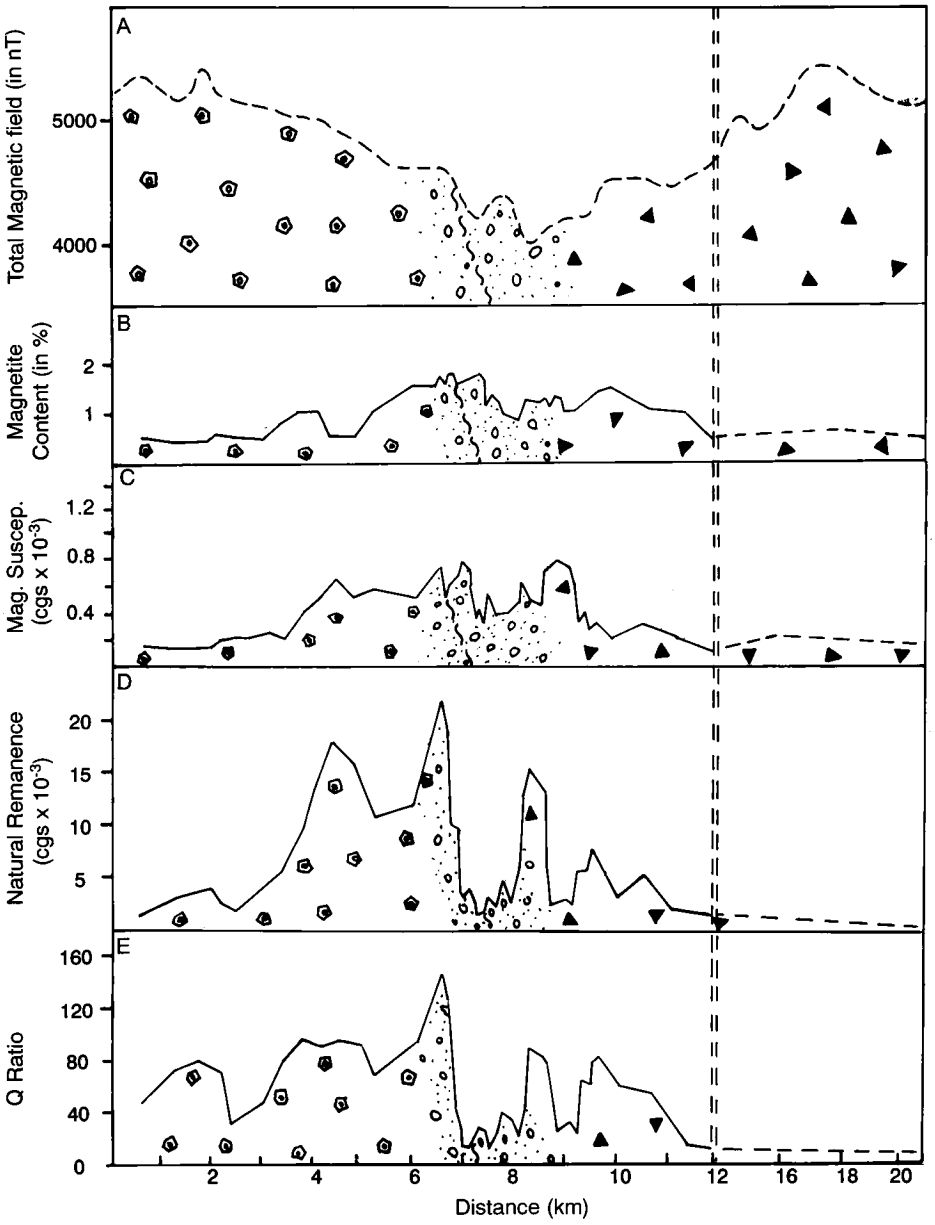


Figure 3. Comparison of profile across airborne magnetic field along traverse A with scalar properties in samples taken along traverse A. Geological symbols are same as in Figure 1. Note that double dashed line indicates change in horizontal (distance) scale. Data for the scalar properties were smoothed by using a three-point moving-average procedure.

and that the remanent magnetization was imprinted on a large volume of the crust below and around the point of impact. We suggest that similar structures may be hidden beneath younger cover rocks elsewhere and that, provided the orientation of the remanence is appropriate, magnetic anomalies detectable even by satellite may be a signature of large (>100-km-diameter) impacts on ancient cratons.

ACKNOWLEDGMENTS

We thank Maarten De Wit and Lew Ashwal for many inspirational discussions over the years, Manfred Hauger and Leonie Maré of the South African Council for Geoscience for doing the susceptibility and remanent measurements of our samples, Ria Tinnion of Geodass (Pty.) Ltd. for processing the airborne magnetic data, and Samual Masala for technical assistance.

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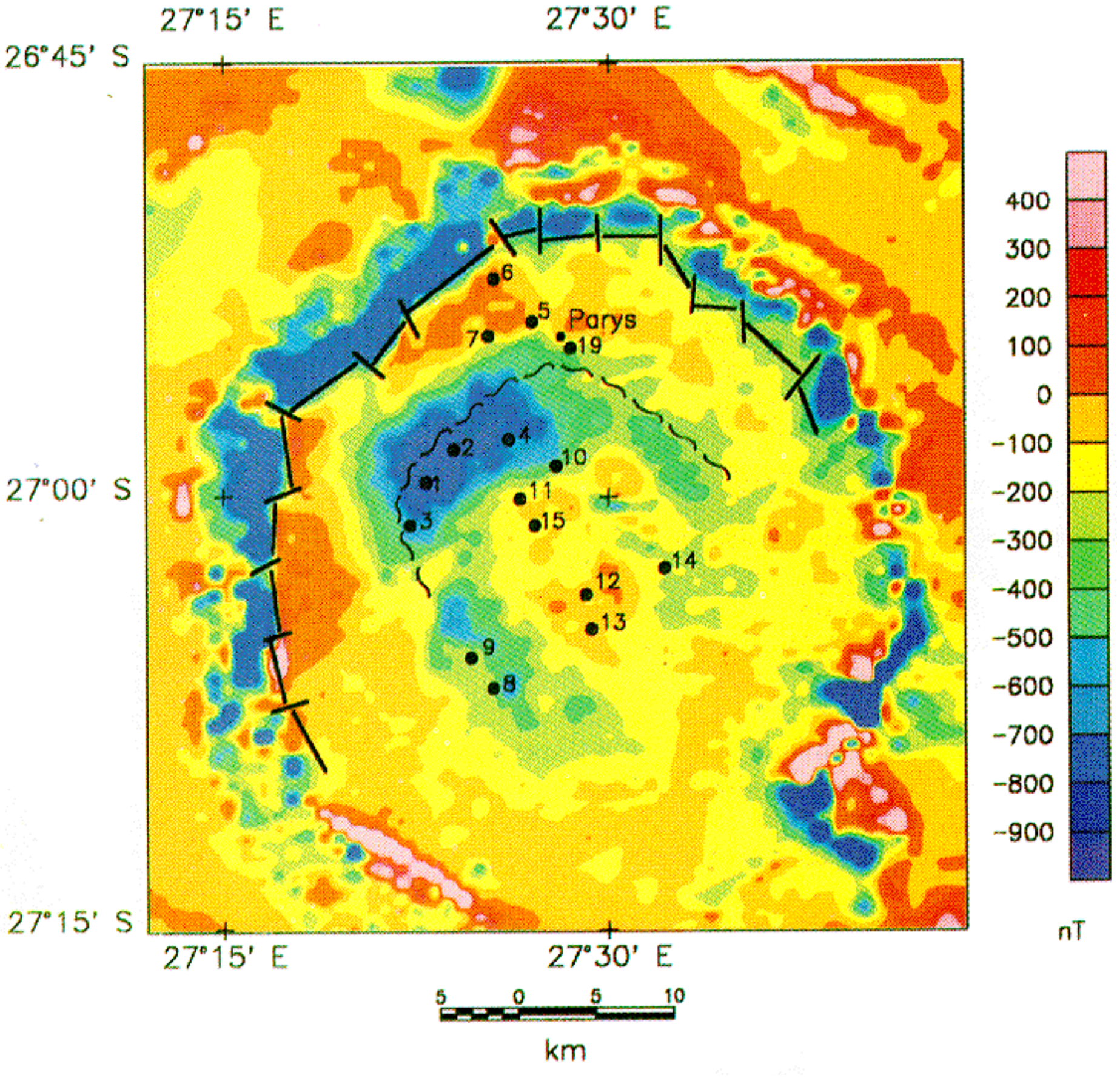
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Manuscript received August 2, 1994
Revised manuscript received October 26, 1994
Manuscript accepted November 11, 1994


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